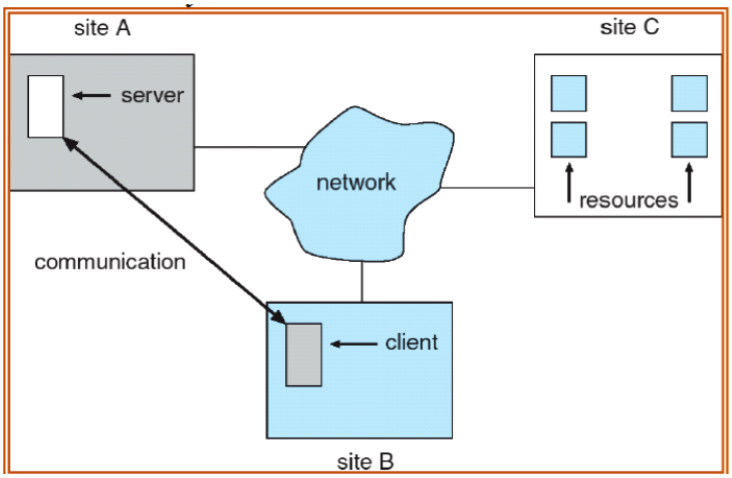
Section (1) :

1. Define Distributed Processing (Computing) Systems

DPS or DCS is : Group of independent computers connected together via a network , providing the view of being a single system.

Examples of DCS :

1. Internet & WWW
2. ATM machines
3. DBMS
4. Mobiles



2. Explain DPS Characteristics.

1. Group of autonomous hosts, each host executes components and operates distribution applications.
2. Hosts are Geographically separated over (LAN, WAN …)
3. Hosts Connected via a network.
4. The network is used to: transfer messages and mail, and execute applications.

Advantages:

1. Resource sharing
2. Openness
3. Scalability
4. Concurrency
5. Fault tolerance

Disadvantages:

1. Complexity
2. Manageability
3. Security
4. Unpredictability

3. Compare between Centralized & Distributed Systems

Advantages of Distribution	Advantages of Centralization
1) Technological changes	1) Economy of scale
2) User needs	2) More powerful capabilities
3) Modularity and simpler software	3) Operating costs
4) Flexibility and extensibility	4) Staff Satisfaction
5) Availability and integrity	5) Local programming and development
6) Performance	6) Communication subsystem
7) Local control	7) Stretching the state of the art

4. Draw and explain Enslow model for distributed processing systems

Enslow Theory: Fully distributed system is :

Fully distributed in Control +

Fully distributed in Data +

Fully distributed in Hardware.

Enslow's Model for Distributed Systems

